Samet Kocbay

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M.Sc. student in Computational Science and Engineering focused on scientific machine learning, PDE-based modeling, and computational physics

Professional experience

- **Max Planck Institute for Plasma Physics [🌐]** *October 2025 - September 2026*
  Working Student / Thesis
  - Develop self-consistent boundary coupling for MHD and electromagnetic interactions in fusion plasmas
  - Implement methods in large-scale nonlinear MHD code (JOREK) for analysis of magnetic island behavior
- **BMW Group [🌐]** *October 2024 - September 2025*
  Working Student
  - Built surrogate model for operator-valued mapping from structural parameters to frequency responses
  - Improved prediction accuracy by >40% using multi-task Gaussian Processes
  - Reduced computational cost by 80% via latent representation learning
- **Compact Dynamics [🌐]** *March 2024 - September 2024*
  Bachelor Thesis
  - Developed an RNN-based thermal neural network for rotor temperature estimation in electric motors, achieving 3–6 °C prediction error
  - Deployed model for real-time inference on embedded control unit, increasing accuracy utilizing EMF corrections
- **KIT - Institute of Fluid Mechanics [🌐]** *July 2023 – December 2023*
  Student Assistant
  - Reconstructed velocity/pressure fields from sparse PIV measurements using PINNs, enabling reduced experimental measurement requirements

Education

- **M.Sc. Computational Science and Engineering** *October 2024 - April 2027*
  Technical University of Munich (TUM), Current Grade: 1.5 (German system)
- **B.Sc. Mechanical Engineering** *October 2019 - September 2024*
  Karlsruhe Institute of Technology (KIT)

Projects

- **KA RaceIng: Head of Aerodynamics** *October 2021 - September 2023*
  - Led 13-person team, over a full aerodynamic development cycle (CFD → manufacturing)
  - Designed and optimized aero package (+14% downforce) using RANS-based CFD (k-ω SST)
  - Performed mesh generation, boundary layer resolution (y+ control), and grid convergence studies
- **ICLR 2026 Challenge: GRaM Workshop [GitHub]** *Currently*
  - Implemented a multiscale neural operator for 3D transient airflow over airfoils
  - Benchmarked temporal strategies (autoregressive, sequence-to-sequence)
  - Achieving preliminary improvements of >9% RL2 reduction vs baseline

Skills

- **Programming:** Python, Linux, Bash, C++, MATLAB, OpenMP, MPI
- **Machine Learning & SciML:** PyTorch, GPyTorch, PINNs, Neural Operators, Gaussian Processes
- **CAE & CFD:** STAR-CCM+, OpenFOAM (RANS, k-ω SST), meshing, turbulence modeling, post-processing, Composite Optimization, FEM, FVM
- **Languages:** German (Native), English (Fluent), Turkish